

Fig. 2: D2M broadcast providing capacity offloading for media content delivery (Source: IIT Kanpur, June 2022 white paper)

Smartphones, laptops, and tablets can consume content without the need for any special app or plugin.

Local content server. A local content server can be hosted to provide offline services.

Receiver for D2M

D2M operates in frequency bands allocated for broadcasting services. Receiving devices must be tunable to these frequency bands to capture signals. The receiver antenna can be integrated into the device or external. The receiver must:

- Decode audio and video content accurately, supporting various codecs and compression techniques
- Handle and correct errors in received signals for a seamless viewing or listening experience
- Support high data throughput to avoid buffering or interruptions
- Prevent unauthorised access or content piracy
- Exhibit low latency for real-time usage

Additionally, to ensure compatibility with broadcasting networks, receivers must meet applicable standards and allow software updates for security, bug fixes, and feature enhancements.

Transmitter for D2M

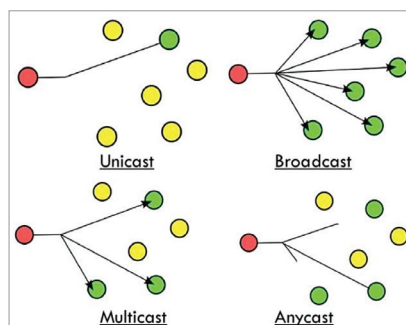
The transmitter must be compatible with receiving devices, operate in the allocated frequency bands, and support the modulation coding scheme. It should also implement

- Handle high data throughput to support high-quality audio and video
- Provide security features to protect content and restrict access to authorised users only
- Be synchronised for seamless handovers and consistent coverage across different cells
- Use antenna systems and power output to effectively cover the intended area

Moreover, the transmitter should be energy-efficient, environmentally friendly, and have low operational costs while adhering to applicable broadcasting standards.

Opposition by industrial stakeholders

D2M technology has promising applications but faces opposition from industry stakeholders, such as telecom service providers, smartphone manufacturers, chipset makers, telecom equipment producers, and network vendors. Stakeholders believe the technology is immature, not ready for large-scale deployment, and lacks



Unicast, multicast and broadcast: different traffic types (Source: <https://orhanergun.net>)

error correction coding techniques to enhance signal robustness.

The transmitter should:

- Exhibit low latency to minimise delay between content generation and transmission

an ecosystem while awaiting global adoption. They argue that further research and development efforts are necessary before effective rollout. Establishing a dedicated D2M network entails substantial infrastructure costs, which stakeholders are reluctant to bear without clear evidence of viability and profitability. Telecom operators are concerned that spectrum allocation for D2M broadcasting infrastructure may disrupt the level playing field. Their apprehension stems from the possibility of D2M offering competing or complementary services, delivering content and data directly to mobile handsets without relying on internet connections and data packs, potentially impacting their revenue streams. Telecom companies have urged the government to ensure D2M services align with national legal and technical standards and avoid regulatory arbitrage.

Smartphone manufacturers are hesitant because current devices do not support D2M technology. To enable compatibility, additional software, baseband processing units, antennas, low-noise amplifiers, and baseband filters must be developed and integrated. This process requires significant investment and time. Manufacturers are also concerned that incorporating broadcasting compatibility could reduce battery performance and cellular reception.

The government has refuted telecom operators' claims of revenue loss, asserting that D2M technology must co-exist with mobile internet or 5G. It has assured device manufacturers of financial support for enabling compatibility. Confident in the technology's maturity, the government anticipates a launch next year following trial completion. Policies and regulations will be finalised only after consulting all stakeholders.

What can consumers expect?

D2M broadcast, primarily aimed at mobility devices, does not support